

FINAL PROJECT REPORT

CUSTOMER CHURN PREDICTION & RETENTION FRAMEWORK

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Certificate

This is to certify that the project report entitled “**Customer Churn Prediction & Retention Framework**” has been successfully completed by **Boda Dhanesh**, Enrollment No. **24162121004**, student of Computer Science & Engineering, Ganpat University.

Signature of Guide

Signature of HOD

Declaration

I, **Boda Dhanesh**, hereby declare that this project report entitled “**Customer Churn Prediction & Retention Framework**” is my original work. The information and analysis presented in this report are based on my learning, implementation, and study of customer churn prediction using Python, Google Colab, and machine learning concepts.

Place: Ganpat University

Date: June 18, 2026

Signature: **Boda Dhanesh**

Acknowledgement

I would like to express my sincere gratitude to my college, faculty members, and project guide for their guidance and support throughout this project. I am thankful to Ganpat University for giving me the opportunity to work on this project.

This project helped me understand Python, Google Colab, data preprocessing, exploratory data analysis, machine learning concepts, and customer churn prediction.

Abstract

Customer churn is a major challenge for businesses because losing customers can reduce revenue and increase the cost of acquiring new customers. The **Customer Churn Prediction & Retention Framework** is a data-driven project designed to identify customers who are likely to stop using a company's products or services.

This project uses the Telco Customer Churn dataset. The dataset contains customer demographic details, service details, billing information, contract type, payment method, tenure, monthly charges, total charges, and churn status.

Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn are used in this project. The work includes data loading, data understanding, data cleaning, handling missing values, duplicate removal, data visualization, and exploratory data analysis. The analysis helps identify important churn factors, such as contract type, internet service, payment method, tenure, and monthly charges.

Contents

1	Introduction	9
1.1	Project Overview	9
1.2	Background of the Study	9
1.3	Problem Statement	9
1.4	Project Objectives	10
1.5	Scope of the Project	10
1.6	Project Deliverables	11
1.7	Report Organization	11
2	Requirement Analysis	12
2.1	Domain Study	12
2.2	Existing System Analysis	12
2.3	Proposed System	12
2.4	Functional Requirements	13
2.5	Non-Functional Requirements	13
2.6	Tools and Technologies Used	14
3	Data Collection and Understanding	15
3.1	Data Source	15
3.2	Data Description	15
3.3	Data Structure	15
3.4	Data Dictionary	16
3.5	Data Quality Assessment	16
4	Data Preprocessing	17
4.1	Data Cleaning	17
4.2	Handling Missing Values	17
4.3	Duplicate Removal	17
4.4	Data Transformation	18
4.5	Data Validation	18
4.6	Final Prepared Dataset	18

5	Data Analysis and Modeling	19
5.1	Exploratory Data Analysis (EDA)	19
5.1.1	Customer Churn Distribution	19
5.1.2	Contract Type Distribution	20
5.1.3	Senior Citizen Distribution	20
5.1.4	Paperless Billing Usage	21
5.1.5	Gender Distribution	22
5.2	Data Relationships	22
5.2.1	Churn Rate by Contract Type	23
5.2.2	Churn Rate by Internet Service	23
5.2.3	Churn Rate by Payment Method	24
5.2.4	Churn Rate by Gender	24
5.2.5	Churn Rate by Customer Tenure	25
5.2.6	Correlation Analysis	25
5.3	Key Metrics Identification	26
5.3.1	Customer Churn Rate	26
5.3.2	Customer Retention Rate	26
5.3.3	Average Monthly Charges	26
5.3.4	Average Customer Tenure	26
5.3.5	Contract-Based Churn Rate	26
5.3.6	Internet Service Churn Rate	26
5.3.7	Payment Method Churn Rate	27
5.3.8	Summary of Key Findings	27

List of Figures

5.1	Enter Caption	19
5.2	Customer Churn Distribution	19
5.3	Enter Caption	20
5.4	Contract Type Distribution	20
5.5	Enter Caption	21
5.6	Senior Citizen Distribution	21
5.7	Enter Caption	21
5.8	Paperless Billing Usage	21
5.9	Enter Caption	22
5.10	Gender Distribution	22
5.11	Enter Caption	23
5.12	Churn Rate by Contract Type	23
5.13	Enter Caption	23
5.14	Churn Rate by Internet Service	23
5.15	Enter Caption	24
5.16	Churn Rate by Payment Method	24
5.17	Churn Rate by Gender	24
5.18	Enter Caption	25
5.19	Churn Rate by Customer Tenure	25
5.20	Enter Caption	25
5.21	Correlation Matrix	25

List of Tables

2.1	Tools and Technologies Used	14
3.1	Dataset Structure	15
3.2	Data Dictionary	16
5.1	Summary of Key Findings	27

Chapter 1

Introduction

1.1 Project Overview

Customer Churn Prediction & Retention Framework is a data-driven system designed to identify customers who are likely to stop using a company's products or services. Customer churn is a major business problem because losing existing customers can reduce company revenue and increase the cost of acquiring new customers.

This project focuses on analyzing customer data to understand churn behavior. The system studies customer features such as tenure, monthly charges, total charges, contract type, internet service, payment method, and churn status. These features help in identifying which customers are more likely to leave the service.

1.2 Background of the Study

In today's competitive business environment, customer retention is very important. Many companies collect customer data, but without proper analysis, it becomes difficult to identify customers who are likely to leave.

Data analytics and machine learning provide a better way to understand customer behavior. By analyzing past customer records, businesses can identify patterns related to churn and take preventive actions.

Customer churn prediction is useful in many domains such as telecommunications, banking, e-commerce, insurance, subscription services, and online platforms.

1.3 Problem Statement

Many organizations face customer loss due to competition, dissatisfaction, high pricing, poor service quality, or changing customer needs. Identifying customers who may leave the service is difficult when the organization has a large amount of customer data.

The problem addressed in this project is to build a Customer Churn Prediction & Retention Framework that can analyze customer data, identify churn patterns, and support businesses in making better retention decisions.

1.4 Project Objectives

The main objectives of this project are:

1. To understand customer churn and customer retention concepts.
2. To collect and study the Telco Customer Churn dataset.
3. To perform data loading and data understanding.
4. To clean the dataset and handle missing values.
5. To remove duplicate records.
6. To perform exploratory data analysis.
7. To identify important factors responsible for churn.
8. To prepare the dataset for machine learning model development.

1.5 Scope of the Project

The scope of this project includes data analysis, data preprocessing, visualization, and churn pattern identification. This framework can be used in:

- Telecommunications
- Banking
- E-Commerce
- Insurance
- Subscription Services
- Online Platforms

1.6 Project Deliverables

The deliverables of this project are:

1. Final project report.
2. Cleaned customer churn dataset.
3. Exploratory data analysis charts.
4. Visual insights and observations.
5. Key churn-related metrics.
6. Prepared dataset for model development.

1.7 Report Organization

This report is divided into different chapters. Chapter 1 introduces the project. Chapter 2 explains requirement analysis. Chapter 3 describes data collection and understanding. Chapter 4 explains data preprocessing. Chapter 5 presents data analysis and modeling up to key metrics identification.

Chapter 2

Requirement Analysis

2.1 Domain Study

Customer churn prediction belongs to the field of data analytics and machine learning. The main purpose of this domain is to analyze customer data and predict whether a customer is likely to continue or discontinue a service.

Businesses use churn prediction to reduce customer loss, improve customer satisfaction, increase revenue, and develop better retention strategies.

2.2 Existing System Analysis

In the existing system, many businesses identify churn only after customers have already left. Manual customer analysis is time-consuming and may not detect hidden patterns in customer behavior.

Limitations of the existing system:

- Manual analysis is slow.
- Hidden churn patterns are difficult to identify.
- Large datasets are difficult to manage manually.
- Retention actions may be delayed.
- Business decisions may not be data-driven.

2.3 Proposed System

The proposed system uses data analysis and machine learning concepts to identify customers who are likely to churn. It analyzes customer attributes such as tenure, monthly charges, contract type, internet service, payment method, and churn status.

Customer Data → Data Preprocessing → EDA → Model Preparation → Churn
Prediction → Retention Strategy

2.4 Functional Requirements

1. The system should load the customer churn dataset.
2. The system should display dataset structure and information.
3. The system should identify missing values.
4. The system should remove duplicate records.
5. The system should convert required columns into proper format.
6. The system should generate visualizations.
7. The system should identify important churn factors.

2.5 Non-Functional Requirements

1. The system should be easy to understand and use.
2. The analysis should be accurate.
3. The code should be readable and well structured.
4. The system should process the dataset efficiently.
5. The project should be scalable for future machine learning implementation.

2.6 Tools and Technologies Used

Table 2.1: Tools and Technologies Used

Tool/Technology	Purpose
Python	Programming language used for data analysis and machine learning
Google Colab	Online platform for writing and running Python code
Pandas	Data loading, cleaning, and analysis
NumPy	Numerical computation
Matplotlib	Data visualization and charts
Seaborn	Advanced data visualization
Scikit-Learn	Machine learning model development
CSV Dataset	Storage format for customer churn data
Google Drive	Dataset storage and notebook access

Chapter 3

Data Collection and Understanding

3.1 Data Source

The dataset used in this project is the Telco Customer Churn dataset. It contains customer information related to telecom services, billing, contract type, service usage, and churn status.

The dataset was loaded in Google Colab using Python and Pandas. The CSV file was read using the `read_csv()` function.

3.2 Data Description

The dataset contains customer details such as gender, senior citizen status, partner, dependents, tenure, phone service, internet service, contract type, paperless billing, payment method, monthly charges, total charges, and churn status.

The target variable is **Churn**, which indicates whether the customer left the service or continued using it.

3.3 Data Structure

The dataset is in tabular format. Each row represents one customer record and each column represents a customer attribute.

Table 3.1: Dataset Structure

Element	Description
Rows	Individual customer records
Columns	Customer features and churn status
File Type	CSV file
Target Variable	Churn
Data Type	Numerical and categorical data

3.4 Data Dictionary

Table 3.2: Data Dictionary

Attribute	Description
customerID	Unique customer identification number
gender	Gender of the customer
SeniorCitizen	Indicates whether the customer is a senior citizen
Partner	Indicates whether the customer has a partner
Dependents	Indicates whether the customer has dependents
tenure	Number of months the customer stayed with the company
PhoneService	Indicates whether the customer has phone service
InternetService	Type of internet service used by the customer
Contract	Type of contract selected by the customer
PaperlessBilling	Indicates whether the customer uses paperless billing
PaymentMethod	Method used by the customer for payment
MonthlyCharges	Monthly amount charged to the customer
TotalCharges	Total amount charged to the customer
Churn	Indicates whether the customer churned or stayed

3.5 Data Quality Assessment

Data quality assessment was performed to check missing values, duplicate records, data types, and consistency.

The following checks were performed:

1. Checked dataset shape and column names.
2. Checked data types using `df.info()`.
3. Checked missing values using `df.isnull().sum()`.
4. Checked duplicate records using `df.duplicated().sum()`.
5. Verified numerical features such as `tenure` and `MonthlyCharges`.

Chapter 4

Data Preprocessing

4.1 Data Cleaning

Data cleaning is an important step before analysis and modeling. In this project, the dataset was checked for incorrect values, missing values, duplicate records, and inconsistent data formats.

The following Python libraries were used:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

4.2 Handling Missing Values

Missing values were checked using:

```
df.isnull().sum()
```

Missing values were handled properly so that the analysis becomes more accurate and reliable.

4.3 Duplicate Removal

Duplicate records were checked using:

```
df.duplicated().sum()
```

Duplicate records were removed using:

```
df = df.drop_duplicates()
```

4.4 Data Transformation

Data transformation was performed to convert data into a proper format. For example, the `TotalCharges` column was converted into numeric format because it is required for numerical analysis.

Categorical variables such as gender, contract type, payment method, and internet service can be encoded before model training.

4.5 Data Validation

After cleaning and transformation, the dataset was validated by checking:

- Dataset shape
- Column names
- Data types
- Missing values
- Duplicate values
- Statistical summary

4.6 Final Prepared Dataset

After preprocessing, the dataset became suitable for exploratory data analysis and machine learning model development. The final prepared dataset contains cleaned customer records with important customer attributes and churn status.

Chapter 5

Data Analysis and Modeling

5.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is one of the most important phases of a data science project. The main objective of EDA is to understand the dataset, identify patterns, detect unusual values, and extract useful insights before applying machine learning algorithms.

In this project, EDA was performed on the Telco Customer Churn dataset using Pandas, Matplotlib, and Seaborn. Different charts were created to understand customer behavior, service usage, billing patterns, and churn trends.

5.1.1 Customer Churn Distribution

Customer churn distribution shows the proportion of customers who stayed with the company and customers who left the service.

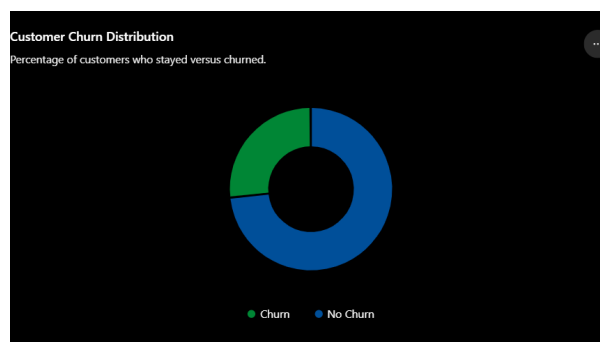


Figure 5.1: Enter Caption

Figure 5.2: Customer Churn Distribution

Observation: Approximately 73.5% of customers remained with the company, while around 26.5% of customers churned.

Inference: Although most customers are retained, the churn percentage is significant and requires attention.

5.1.2 Contract Type Distribution

Contract type is an important factor in customer retention. It shows how customers are distributed among Month-to-Month, One-Year, and Two-Year contracts.

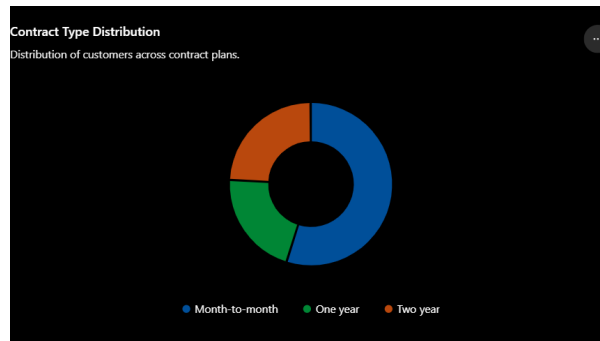


Figure 5.3: Enter Caption

Figure 5.4: Contract Type Distribution

Observation: Most customers prefer Month-to-Month contracts, followed by Two-Year and One-Year contracts.

Inference: Customers prefer flexible contract options, but flexible contracts are also associated with higher churn.

5.1.3 Senior Citizen Distribution

This analysis shows the distribution of senior citizen and non-senior citizen customers.

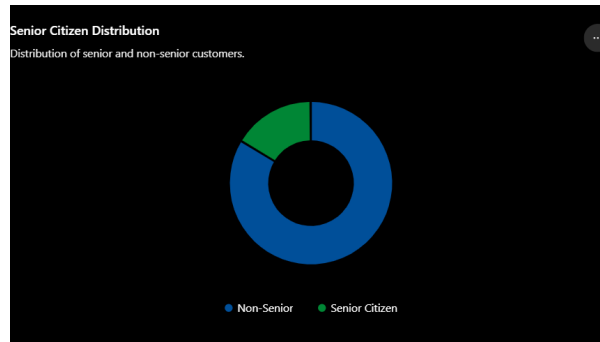


Figure 5.5: Enter Caption

Figure 5.6: Senior Citizen Distribution

Observation: Senior citizens represent a smaller portion of the customer base.

Inference: Although senior citizens are fewer in number, they may show higher churn behavior.

5.1.4 Paperless Billing Usage

Paperless billing analysis shows how many customers use digital billing compared to traditional billing.

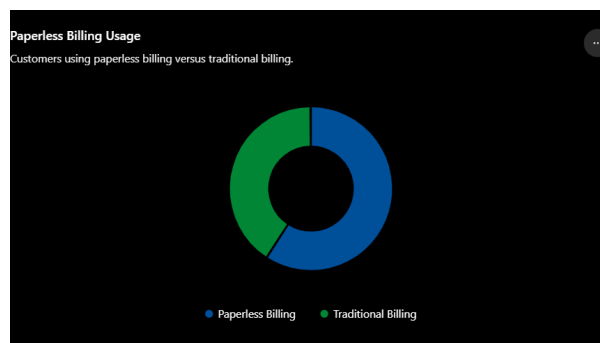


Figure 5.7: Enter Caption

Figure 5.8: Paperless Billing Usage

Observation: Most customers have adopted paperless billing.

Inference: Digital billing is widely accepted among customers and reflects modern service preferences.

5.1.5 Gender Distribution

Gender distribution was analyzed to check whether customer gender affects churn behavior.

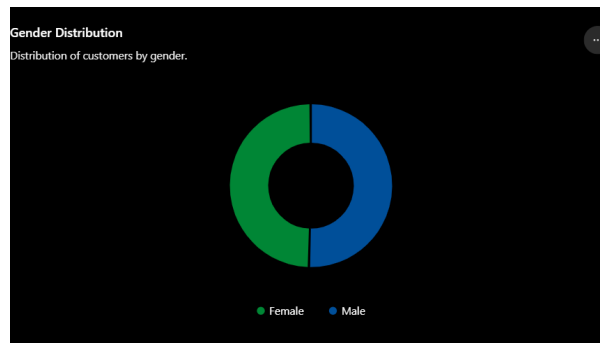


Figure 5.9: Enter Caption

Figure 5.10: Gender Distribution

Observation: The customer base is nearly equally divided between male and female customers.

Inference: Gender does not significantly influence customer enrollment or churn behavior.

5.2 Data Relationships

Data relationship analysis helps identify how different customer attributes affect churn.

5.2.1 Churn Rate by Contract Type

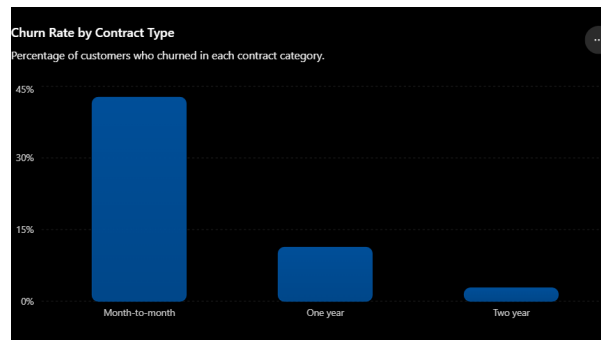


Figure 5.11: Enter Caption

Figure 5.12: Churn Rate by Contract Type

Observation: Month-to-Month customers have the highest churn rate, while Two-Year contract customers have the lowest churn rate.

Inference: Long-term contracts contribute significantly to customer retention.

5.2.2 Churn Rate by Internet Service

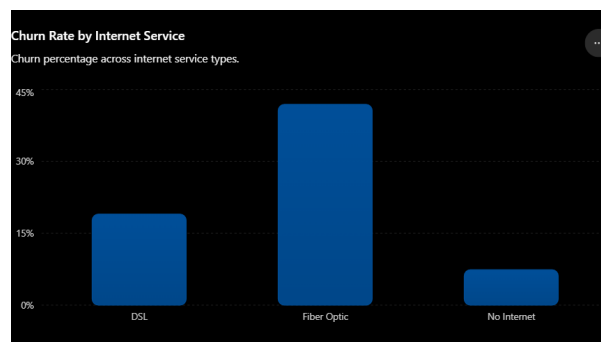


Figure 5.13: Enter Caption

Figure 5.14: Churn Rate by Internet Service

Observation: Fiber Optic users have the highest churn rate.

Inference: Service quality, pricing, or customer expectations may influence churn among Fiber Optic users.

5.2.3 Churn Rate by Payment Method

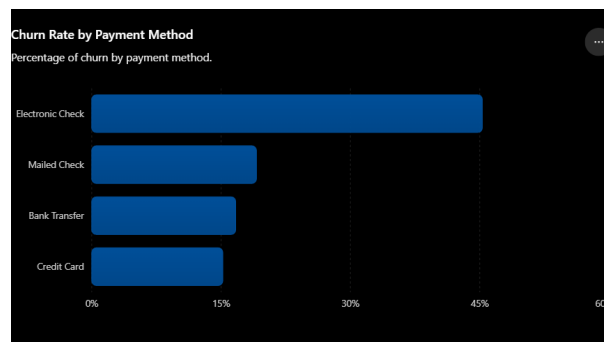


Figure 5.15: Enter Caption

Figure 5.16: Churn Rate by Payment Method

Observation: Electronic Check users show the highest churn percentage.

Inference: Customers using Electronic Check payments may be less satisfied or less committed to the service.

5.2.4 Churn Rate by Gender

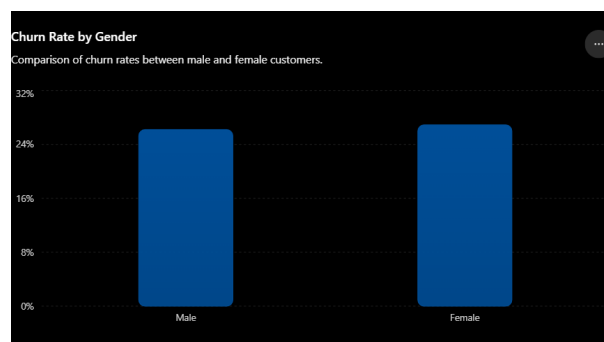


Figure 5.17: Churn Rate by Gender

Observation: Male and female customers show nearly identical churn rates.

Inference: Gender is not a major factor affecting customer churn.

5.2.5 Churn Rate by Customer Tenure

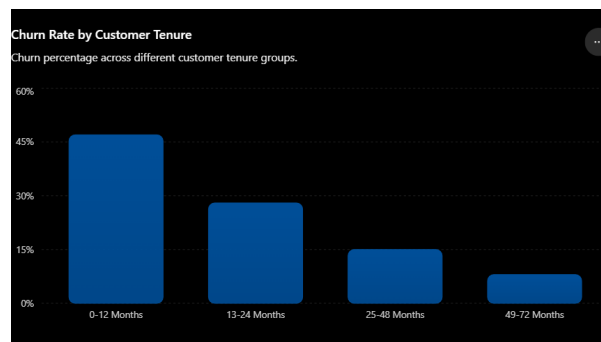


Figure 5.18: Enter Caption

Figure 5.19: Churn Rate by Customer Tenure

Observation: Customers within the first year have the highest churn rate.

Inference: Customer loyalty increases as the duration of service increases.

5.2.6 Correlation Analysis

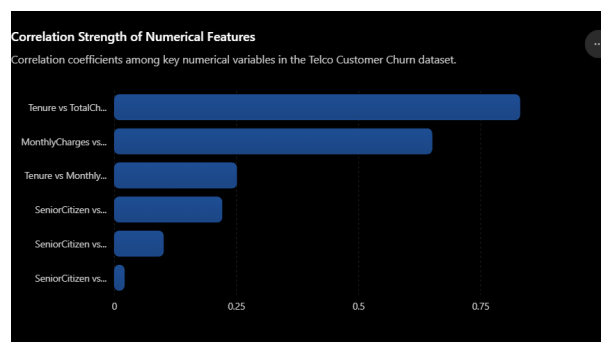


Figure 5.20: Enter Caption

Figure 5.21: Correlation Matrix

Observation: Numerical variables such as tenure, MonthlyCharges, and TotalCharges show useful relationships.

Inference: Tenure is an important factor because customers who stay longer are less likely to churn.

5.3 Key Metrics Identification

Key metrics are important for understanding customer churn and business performance.

5.3.1 Customer Churn Rate

Churn rate measures the percentage of customers who left the service.

$$\text{Churn Rate} = \frac{\text{Number of Churned Customers}}{\text{Total Customers}} \times 100$$

5.3.2 Customer Retention Rate

Retention rate measures the percentage of customers who continued using the service.

$$\text{Retention Rate} = \frac{\text{Number of Retained Customers}}{\text{Total Customers}} \times 100$$

5.3.3 Average Monthly Charges

Average monthly charges help analyze customer spending patterns.

$$\text{Average Monthly Charges} = \frac{\sum \text{Monthly Charges}}{\text{Total Customers}}$$

5.3.4 Average Customer Tenure

Average tenure indicates how long customers stay with the company.

$$\text{Average Tenure} = \frac{\sum \text{Tenure}}{\text{Total Customers}}$$

5.3.5 Contract-Based Churn Rate

Contract-based churn rate evaluates churn percentage for each contract type:

- Month-to-Month
- One-Year
- Two-Year

5.3.6 Internet Service Churn Rate

This metric identifies churn trends across internet service types:

- DSL

- Fiber Optic
- No Internet Service

5.3.7 Payment Method Churn Rate

This metric compares churn across payment methods:

- Electronic Check
- Mailed Check
- Bank Transfer
- Credit Card

5.3.8 Summary of Key Findings

Table 5.1: Summary of Key Findings

Metric	Finding
Churn Rate	Around 26.5%
Retention Rate	Around 73.5%
Highest Churn Contract	Month-to-Month
Lowest Churn Contract	Two-Year
Highest Churn Internet Service	Fiber Optic
Highest Churn Payment Method	Electronic Check
Customer Loyalty Indicator	Tenure
Gender Impact	Minimal